



Lucas Lemos Ricaldoni

Date of birth: 14/10/2007 | **Nationality:** Italian, Brazilian | **Gender:** Male | **Phone:** (+55) 31997091410 (Mobile) | **Email address:** lucaslemosricaldoni@gmail.com | **Address:** Belo Horizonte, Brazil (Home)

EDUCATION & TRAINING

31/12/2022 - 31/12/2025 - BELO HORIZONTE, BRAZIL

HIGH SCHOOL DIPLOMA WITH PROFESSIONAL QUALIFICATION IN INDUSTRIAL AUTOMATION-TECHNICAL COLLEGE OF UFMG

Technical training focused on **Industrial Automation**, **Robotics**, and **Embedded Systems**. **Electronics and Programming:** Developing circuits and programming in C/C++ and Python for microcontrollers like Arduino and ESP32. **Robotics:** Designing and programming autonomous line-follower robots using PID control. **3D Modeling:** Expert use of SolidWorks and Fusion 360 for mechanical design and prototyping. **Industrial Control:** Knowledge of PLC programming (Ladder, ST, SFC) and SCADA systems

Level in EQF: 4

WORK EXPERIENCE

28/02/2025 - 31/01/2026 - BELO HORIZONTE, BRAZIL

RESEARCH AND PROTOTYPING INTERN LIPS - INNOVATION, PROTOTYPING AND SIMULATION LABORATORY (UFMG)

* Developing innovative technological solutions focusing on robotics and automation. * Advanced 3D modeling and mechanical design using SolidWorks. * Programming embedded systems and microcontrollers in C/C++. * Prototyping, assembly, and testing of electronic circuits and robotic hardware.

SKILLS

Programming Languages

Python (computer programming) | C | C++ | MATLAB | JavaScript

Engineering & Modeling

robotics | Fusion 360 | Matlab Environment (Matlab, Simulink, Simscape) | SolidWorks | Creality Print | use CAD software | printing techniques | 3D printing process

Development Tools

Microsoft Excel | LaTeX | Git & GitHub | Arduino IDE & Visual Studio Code

LANGUAGE SKILLS

Mother tongue(s): **PORTUGUESE**

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ITALIAN	B2	B2	B2	B2	B2

ENGLISH	B2	B2	B1	B1	B1
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● HONOURS AND AWARDS

18/09/2025 Federal University of Minas Gerais (UFMG)

3rd Place (Advanced Line Follower) - CoRA 2025 As Captain of the "Tartaruga Linhas" team, led the development of autonomous robots using Arduino and C/C++. Won 3rd place in the competitive advanced line-follower category.

Links

<https://www.linkedin.com/in/lucaslemosricaldoni/overlay/Honor/577107598/treasury/?profileId=ACoAAE0KWKsBz5pJWhSXUIABoaaO85RukP9AKWg>

18/09/2025 Federal University of Minas Gerais (UFMG)

1st Place - Best Design CoRA 2025 As Captain of the "Tartaruga Linhas" team, led the development of autonomous robots using Arduino and C/C++. Won 1st place for technical design.

Links

<https://www.linkedin.com/in/lucaslemosricaldoni/overlay/Honor/577187033/treasury/?profileId=ACoAAE0KWKsBz5pJWhSXUIABoaaO85RukP9AKWg>

01/10/2025 Federal University of Minas Gerais (UFMG)

Academic Relevance Award - UFMG Knowledge Week 2025 Recognized for outstanding academic contribution and research quality during the annual UFMG scientific showcase. Developed a complete IoT system designed for elderly care, featuring a smart device that monitors plant humidity and sends automated alerts to ensure a healthy environment for the user

Links

<https://www.linkedin.com/in/lucaslemosricaldoni/overlay/Honor/242643532/treasury/?profileId=ACoAAE0KWKsBz5pJWhSXUIABoaaO85RukP9AKWg>

26/09/2025 Center for Artificial Intelligence in Health (CI-IA)

1st Place - Geraldo Ribeiro Award | CI-IA Health Challenge Developed the winning 3D modeling and mechanical simulations for an innovative hospital screening technological solution.

Links

<https://www.linkedin.com/in/lucaslemosricaldoni/overlay/Honor/1272134977/treasury/?profileId=ACoAAE0KWKsBz5pJWhSXUIABoaaO85RukP9AKWg>

● CERTIFICATIONS

31/12/2025 Harvard University

Aplicativos de TinyML

Online - Professional certification covering the deployment of machine learning models on low-power, resource-constrained microcontrollers (24 hours)

31/12/2025 Harvard University

CS50's Web Programming with Python and JavaScript

Project Based - > Comprehensive course on designing and implementing web apps with Python, JavaScript, and SQL using frameworks like Django and React (80 hours).

Links <https://certificates.cs50.io/a1ed26eb-30f0-4de8-ac4a-2c8319979ea3.pdf?size=letter>

31/12/2025 MathWorks

MathWorks Certification - Simulink, Simscape, and Multibody Simulation

Project Based - Technical training in multi-domain physical modeling and simulation for mechanical and

robotic systems (10 hours).

31/12/2025 OpenCV University

OpenCV Bootcamp

Project Based - Practical certification focused on image processing, object detection, and computer vision algorithms (10 hours).

Links <https://courses.opencv.org/certificates/289e1aae3c0f4bf1b3a47d8f19d73e47>

31/12/2025 Federal Institute of Minas Gerais (IFMG)

Industry 4.0: Internet of Things

Online - Focused on Industrial Internet of Things (IIoT) and general IoT concepts, covering sensors, connectivity, and data application in industrial environments (30 hours).

Links

<https://www.linkedin.com/in/lucaslemosricaldoni/overlay/Certifications/1730704889/treasury/?profileId=ACoAAE0KWKsBz5pJWhSXUIABoaaO85RukP9AKWg>